Name: Marking

Teacher: Key

Class: ____________________

CC number: ____________________

Baldivis
Secondary College

MATHEMATICS
YEAR 9 EXAM
SEMESTER ONE 2015

Section One: Resource-Free

TIME ALLOWED FOR THIS PAPER

Reading time before commencing 2 minutes.
Working time for paper 33 minutes.

MATERIAL(S) REQUIRED / RECOMMENDED FOR THIS PAPER

TO BE PROVIDED BY THE SUPERVISOR:
This Question / Answer Booklet

TO BE PROVIDED BY THE CANDIDATE:
Standard items: pens, pencils, pencil sharpener, eraser, correction fluid, ruler, highlighters.

Special items: nil

Important note to candidates: No other items may be used in this section of the examination. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Instructions to candidates

- Candidates are to remain seated during the exam and are not allowed to talk to other candidates.
- Mobile phones are not to be used at all during the exam.
- Students who fail to follow examination rules will be removed and dealt with in accordance to the schools policy.

Write your answers in the spaces provided in this Question/Answer Booklet. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use spare pages for planning, indicate this clearly at the top of the page
- Continuing an answer: If you need the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Show all your working clearly. Your working should be sufficient to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

Use a blue or black pen only, except with diagrams where you should use pencil.

Question	*Out of*	*Mark*
1	1	
2	1	
3	1	
4	1	
5	1	
6	1	
7	1	
8	1	
9	1	
10	1	
11	1	
12	1	
13	1	
14	1	
15	1	
16	5	
17	4	
18	5	
19	6	
TOTAL	***35***	
	Mark	
	Percentage	

1. $2^{12} \div 2^{7}$ is equal to: $2^5 = 32.$

A 4 B 8 C 16

D 25 E 32

[1 mark]

2. $\left(\frac{4}{9}\right)^2$ is equal to:

A $\frac{4}{81}$ B $\frac{8}{18}$ C $\frac{16}{9}$

D $\frac{16}{81}$ E $\frac{16}{108}$

[1 mark]

3. $(3x)^3$ in expanded form is: (circle the correct answer)

A $x \times x \times x$ B $3 \times x \times x \times x$ C $3 \times 3 \times 3 \times x \times x \times x$

D $3x^3 \times 3x^3 \times 3x^3$ E $(3x)^3 \times (3x)^3 \times (3x)^3$

[1 mark]

4. $2x^7 \times 7x^3$ is equivalent to:

A $14x^{21}$ B $14x^{10}$ C $14x^{4}$

a. D $9x^{10}$ E $9x^{21}$

[1 mark]

5. $\frac{6a^7b^5}{3a^2b^8}$ simplifies to:

A $\frac{2a^5}{b^3}$ B $\frac{b^5}{2a^3}$ C $\frac{2a^3}{b^5}$

D $\frac{2b^3}{a^5}$ E $\frac{a^5}{2b^3}$

[1 mark]

6. $-2x^0 + 9x^0$ simplifies to:

A -2 B -1 C 0

D 2 E 7

[1 mark]

7. The number of significant figures given in 0.07309 is:

A two B three C four

D five E six

[1 mark]

Questions **8** and **9** refer to the following expression: $2y + 5xy - 3x - 7$.

8. The coefficient of x is:

A 2 B 3 C 5

[1 mark]

D -3 E -7

9. The constant term is:

A $2y$ B $-3x$ C -7

D 7 E $5xy$

[1 mark]

10. $3y + 7y - y$ simplifies to:

A $8y$ B $9y$ C $10y + 1$

D 9 E $10y$

[1 mark]

11. When we expand brackets, we use the distributive law.
Which of the following does *not* represent the distributive law?

A $a(b + c) = ab + ac$ B $-a(b + c) = -ab - ac$ C $-a(b - c) = -ab - ac$

D $-a(b - c) = -ab + ac$ E $a(b - c) = ab - ac$

[1 mark]

12. The highest common factor of $18x^2yz$ and $24y^2z$ is:

A $6yz$ B $3xyz$ C $3x^2y$

D $4y^2z$ E $6xy^2z$

[1 mark]

13. When $4a + 8ab$ is fully factorised, the result is:

A $4(a + 2b)$ **B** $2a(2 + 4b)$ **C** $4a(1 + 2b)$

D $4(a + 8ab)$ **E** $2a(4 + 8b)$

[1 mark]

14. $(3x-2)(x-4)$ in expanded and simplified form is:

A $3x^2+10x-8$ **B** $3x^2-14x+8$ **C** $3x^2-6x+8$

D $3x^2+14x-8$ **E** $3x^2-10x+8$

[1 mark]

15. Which of the following mathematical statements is *true*?

A $(a-b)^2 = a^2+2ab-b^2$ **B** $(a-b)(a+b) = a^2-2ab+b^2$

C $(a+b)(a-b) = a^2+b^2$ **D** $(a+b)^2 = a^2+2ab+b^2$

E $(a-b)^2 = a^2-2ab-b^2$

[1 mark]

16. Evaluate:

a. $4^3 =$ 64.

b. $12^1 =$ 12

c. $9^0 =$ 1

d. $2^{-3} =$ $\frac{1}{2^3} = \frac{1}{8}$

e. $\frac{1}{5^{-2}} =$ $5^2 = 25$

[5 mark]

17. Match the keyword with the definition

Variable	The numbers being multiplied by the variables
Coefficient	The combination of letters,numbers and operation
Expression	The term without a variable
Constant term	The letter used to represent numbers

[4 Marks]

18.

a. Write 1.86×10^2 as a decimal number.

186·0

b. Write 410 500 000 in scientific notation.

$4{\cdot}105\times10^{8}$

c. Write 0.000 096 in scientific notation.

$9{\cdot}6\times10^{-5}$

[5 marks]

19.

	A	not A	Total
B	9	5	14
not B	4	2	6
Total	13	7	20

✓✓

a. Fill in the missing values in the table above.

b. Use the table to find:

i. Pr(A) $\frac{13}{20}$ ✓

ii. PR(not A) $\frac{7}{20}$ ✓

iii. Pr(A and B) $\frac{9}{20}$ ✓

iv. Pr(A or B) $\frac{9}{20}$ ✓

[6 marks]

[End of Section One]

Additional working space